Selective catalytic reduction of NO_x on gasborne catalyst particles

J. Beimdiek, S. Schiller, H.-J. Schmid, Paderborn University, Paderborn/Germany

The exhaust gas of combustion processes – particularly for biomass combustion – typically contains a significant number of hazardous particles and gas components. Therefore, emission legislation has become increasingly restrictive regarding these pollutants, particularly NO_x. The prevailing technology for exhaust gas denitrification is selective catalytic reduction (SCR), which involves the catalytically enabled conversion of harmful NO_x with ammonia to the harmless species nitrogen and water. Commonly, this is done using vanadia/titania catalysts, which are e.g. processed in filter bags or installed as standalone catalyst modules. An alternative SCR technology is currently being investigated and presented here.

The prerequisite is that ultrafine particle separation is ensured. Many industrial filtration systems use a precoat material applied to the filter surface. Such a system efficiently precipitates even ultrafine particles and simultaneously prevents clogging of the filter material. Hence, the addition of further ultrafine particles is harmless. Therefore, the new SCR technique involves the addition of catalytically active nanoparticles synthesized in-situ in the exhaust gas stream. SCR occurs in-flight at the freshly created particle surface. Downstream, the catalyst particles are precipitated with the fly ash particles in the surface filter. Because the nanoparticles are still reactive, the filter cake serves as a fixed bed catalyst and therefore enhances the NO_x separation performance. In contrast to the precoat material, the catalyst particles are not recycled. Hence, the catalyst material must be inexpensive. This represents an incentive to replace established catalysts such as vanadium oxide by e.g., iron oxide.

The integration of a particle synthesis step into the exhaust gas system has several advantages over other SCR systems. First, nanoparticles exhibit a high specific surface area. Therefore, the consumption of catalyst material is low. Second, nanoparticles have a highly reactive surface. This enables either the use of less active metal oxides (as e.g. iron oxide) as catalyst material or the reduction of the operating temperature for SCR. Low operating temperatures are desirable and especially beneficial for combustion processes coupled with heat recovery.

Experimental results with iron oxide particles prove the efficacy of the presented novel SCR concept. However, some challenges remain. The flame-synthesized

particles contain significant amounts of carbonaceous impurities. Their reduction is important for improving SCR performance. In addition, the increased reactivity of the catalyst not only favors SCR but also ammonia oxidation (SCO). Because ammonia is needed to convert nitrogen oxides, this must be prevented. Further, for optimal SCR performance, a catalyst should possess a balanced ratio of acid and redox sites, which influence the adsorption as well as the activation of NH_3 and NO_x for the formation of N_2 and H_2O. Pure iron oxide e.g., exhibits high redox ability but poor acidity. Composites of iron oxide and manganese oxide, however, exhibit a high acid-redox synergy and therefore are currently the focus of investigation.

The SCR performance of the composite particles as well as the influence of carbon residues and particle properties on SCR efficiency will be presented for this novel SCR°technique which is a promising approach and has the potential to become an attractive upgrade measure for exhaust gas cleaning systems with an existing precoat filtration stage.